

Capstone presentation deck

SLIDE 01 / 08

ControlSift

Can a small language model help under-resourced teams tell proof from paperwork?

- MMC · DeepMind AI Research Foundations capstone
- Official SDG: **Goal 10 — Reduced Inequalities**

Problem

- GRC expertise is unequally distributed
- Policy PDFs get treated as operational proof
- False assurance hits smaller organizations hardest

Research aim

- Five-class evidence quality on a sealed synthetic benchmark
- Compare classical → prompted Gemma → QLoRA
- Primary metric: macro F1 · seed 42 · human judgment remains final

Method

- 1,500 compositional packets · family-isolated splits
- Protocol sealed before LLM claims
- Gemma 3 1B IT + QLoRA on free-tier GPU path

Results so far

- Majority macro F1: 0.067
- TF-IDF: 0.533 test / 0.524 challenge
- Gemma zero-shot: 0.080
- Gemma few-shot: 0.137
- Gemma QLoRA: 0.083

Responsible innovation

- Synthetic data only · Data Card · Risk Register
- No production-auditor claims
- Failure Lab documents hard cases

Impact (SDG 10)

- Open methods reduce dependence on expensive GRC consulting
- Free-tier reproduction keeps access wide
- Triage aid — not automated certification

Next steps & ask

- Gemma TEST metrics published from result files
- Publish metrics from files; refresh Failure Lab
- Artifacts: research site · report · Assurance · curriculum map

